

1. Let $e^{-4t} \cos(3t) = \mathcal{R}\{e^{st}\}$, which ones of the followings is the value of s ?

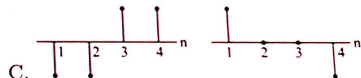
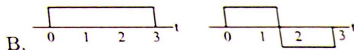
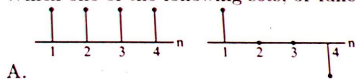
A. -4

B. 3

C. $-4 + 3j$

D. $-4 + (3 + \frac{\pi}{2})j$

2. Which one of the following sets, or functions are not orthogonal to each other?



D. $\{e^{j\frac{2\pi}{13}n}\}, \{e^{j\frac{5\pi}{13}n}\}$

3. Which one of the followings is not an LTI system?

A. $y[n] = \sum_{k=n-3}^n x[k] + 3$

B. $y(t) = \int_{-3}^0 x(t-\tau)e^{\tau} d\tau$

C. $y(t) = \int_{t-3}^t x(\tau)e^{t-\tau} d\tau$

D. $y[n] = \sum_{k=-\infty}^{\infty} x[k] \frac{1}{(n-k)^2}$

4. Which one of the following signals is aperiodic?

A. $x(t) = \sin(3t) + \cos(10t + \frac{\pi}{6})$

B. $x[n] = \sin(3n) + \cos(10\pi n)$

C. $x[n] = e^{j\frac{\pi}{2}n} + e^{j\frac{\pi}{4}n}$

D. $x(t) = 2 \cos(\frac{n\pi}{4}) + 3 \cos(\frac{n\pi}{2})$

5. Which one of the following systems is not linear?

A. $y[n] = (n+6)x[n]$

B. $y[n] = \sum_{k=0}^5 x[n-k]$

C. $y[n] = x[n] \cdot x[n-3]$

D. $y[n] = x[2n]$



6. Which of the signals specified below have Fourier series?

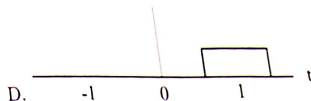
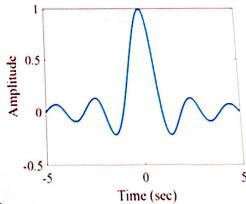
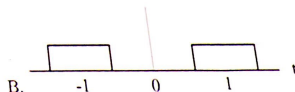
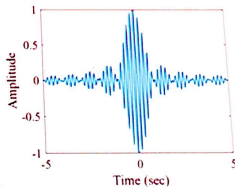
A. $x(t) = e^{2t}u(t)$

B. $x(t) = e^t + e^{-t}$

C. $FT\{x(t)\} = \sin(\omega)$

D. $FT\{x(t)\} = \delta(\omega - 2) * \delta(\omega - 4)$

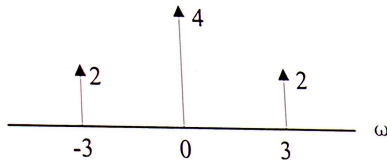
7. Which one of the following impulse responses is a bandpass filter?



8. A CT system $h(t)$ has frequency response of $H(j\omega) = e^{-j3\omega}$, which one of the following statements is false?

- A. The magnitude response is the same at all frequency
- B. This is a linear phase system
- C. This is a casual system
- D. This is not a stable system

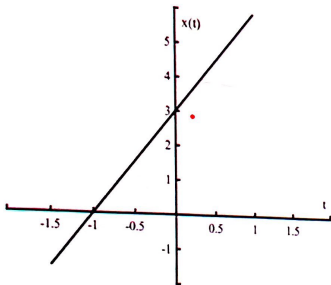
9. The spectrum of a signal is shown below, which one of the followings is false?



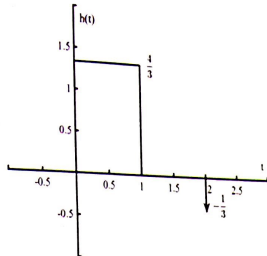
- A. This signal is a CT signal
C. This signal is a periodic signal
B. This signal has a DC term
D. This signal is a DT signal
10. Which one of the following systems is not time invariant?
- A. $y(t) = x(t + 0.3)$
C. $y(t) = \sin(x(t + 0.2))$
B. $y(t) = x(t)e^{-\sqrt{2}t}$
D. $y(t) = x(t - 1.4) + x(t + 1.4)$

Question 11 (10')

For the pair of waveforms shown in the figures below, use the convolution integral to find the response $y(t)$ of the LTI system with impulse response $h(t)$ to the input $x(t) = 3t + 3$. Plot the waveform of $y(t)$.



Q.11 figure of $x(t)$



Q.11 figure of $h(t)$

Question 12 ($4' + 6' = 10'$)

- a) Classify the following signals as periodic or aperiodic and find the fundamental period, if periodic:
- i) $x(t) = \cos(2t + \frac{2\pi}{3})^2$
 - ii) $x[n] = \cos(\frac{\pi}{4}n) + \sin(\frac{\pi}{8}n) - \cos(\frac{\pi}{2}n + \frac{\pi}{3})$
- b)
- i) For a continuous signal $x(t) = 0.5e^{-2t}u(t)$, determine the total energy and the average power of the signal.
 - ii) State (without proof), whether the system with input-output relation $y(t) = \int_{-\infty}^t x(\tau)d\tau$, possess the the following properties: invertibility, linearity, time invariance, memoryless, causality.

Question 13 ($5' + 5' + 5' = 15'$)

For a discrete-time periodic signal $x[n] = \sin(\frac{\pi n}{4}) + 2 \cos(\frac{\pi n}{2})$,

- a) What is its period value (N)?
- b) Determine its non-zero Fourier series coefficients from a_{-5} to a_3 .
- c) What are the values of its Fourier series coefficients a_{20} and a_{2022} ?

Question 14 (15')

Suppose we are given the following information about a continuous-time periodic signal with period 3 and Fourier coefficients a_k :

(1) $a_k = a_{k+2}$

(2) $a_k = a_{-k}$

(3) $\int_{-0.5}^{0.5} x(t) dt = 1$

(4) $\int_{0.5}^{1.5} x(t) dt = 2$

Determine $x(t)$.

Question 15 ($5' + 5' + 10' = 20'$)

A casual and stable LTI system S has the frequency response

$$H(j\omega) = \frac{3j\omega + 1}{2 - \omega^2 + 3j\omega}$$

- Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S .
- Determine the impulse response $h(t)$ of S .
- What is the output of S when the input is $x(t) = \frac{1}{3}e^{-\frac{1}{3}t}u(t) + \frac{2}{9}te^{-\frac{1}{3}t}u(t)$?